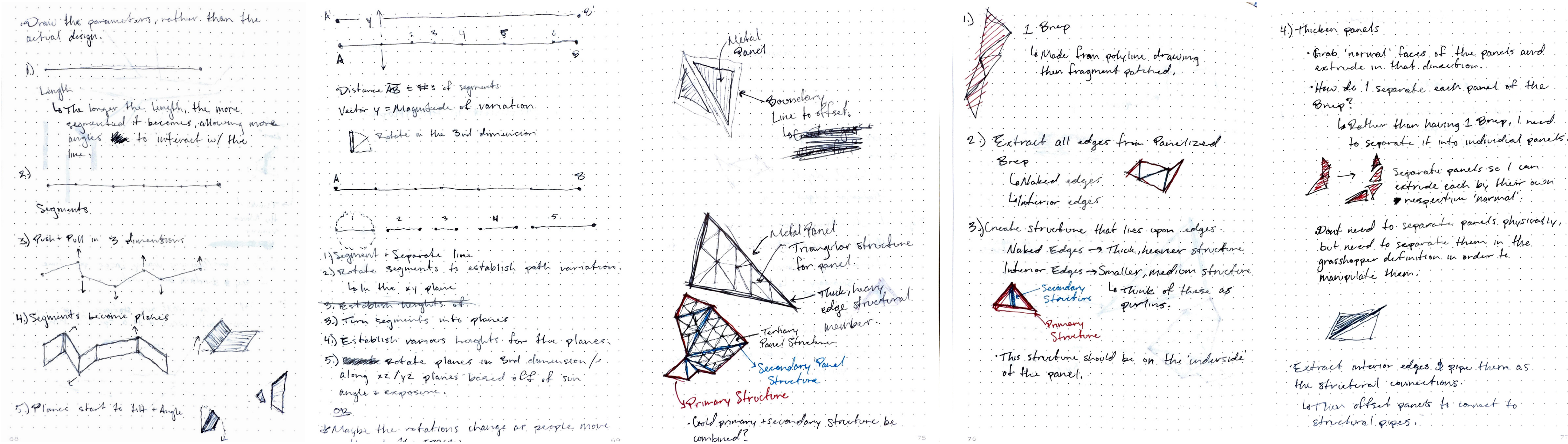
assignment 02

parametric pavilion

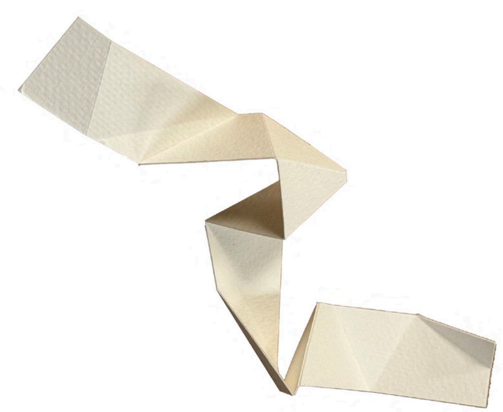
gage carlton

arch565
advanced computer applications

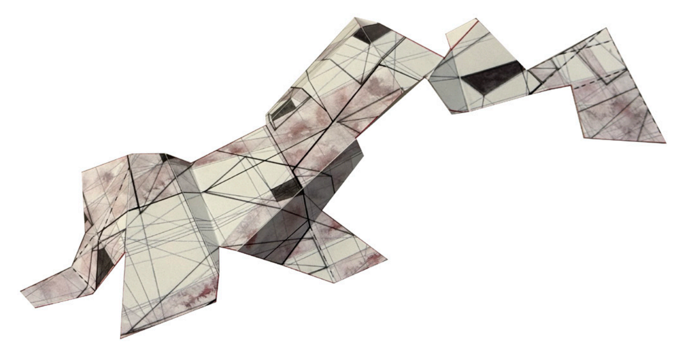


parametric sketching + idea forming

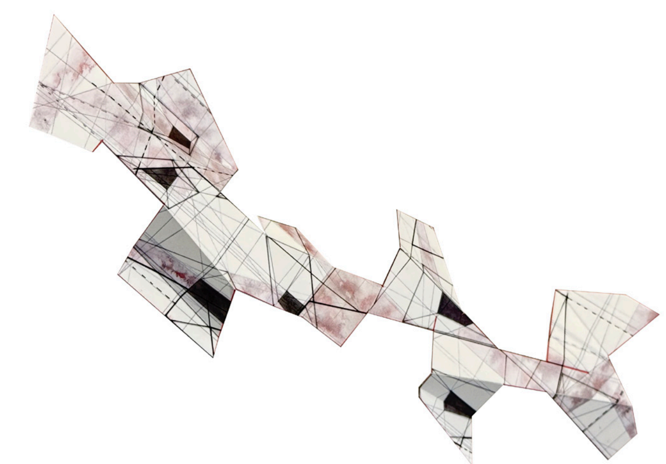
In my advanced graduate studio, I have been exploring how space and form can be created from a series of cuts, folds, and rotations. These few models were a study into how both expected and unexpected ways of folding could create a pavilion-like structure. I thought this assignment provided a perfect chance to explore ways to create and develop these forms through the use of physical modeling, sketching, Grasshopper, Revit, and Twinmotion workflows.



study model 1 | folded watercolor paper



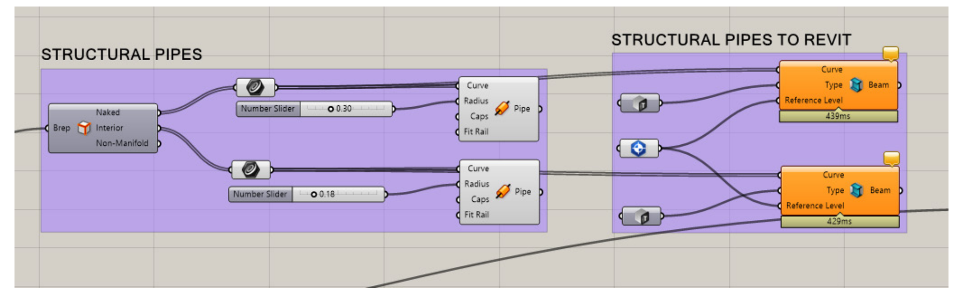
study model 1 | folded cardstock



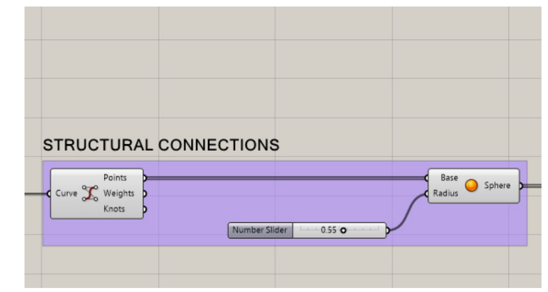
study model 3 | folded cardstock

sketches + study models

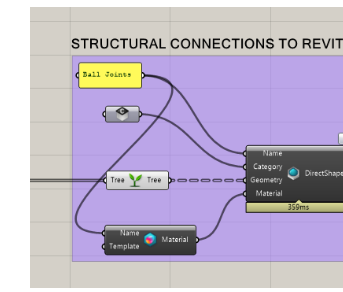
3. structural components



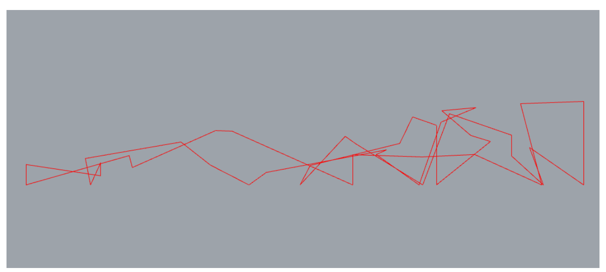
4. structural joints



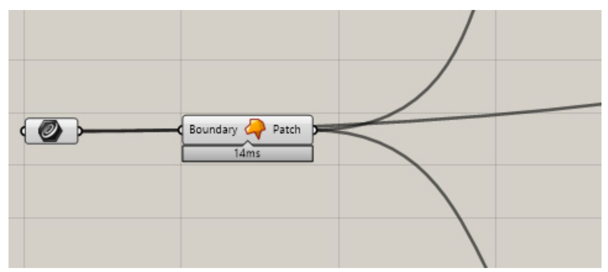
5. rhino inside



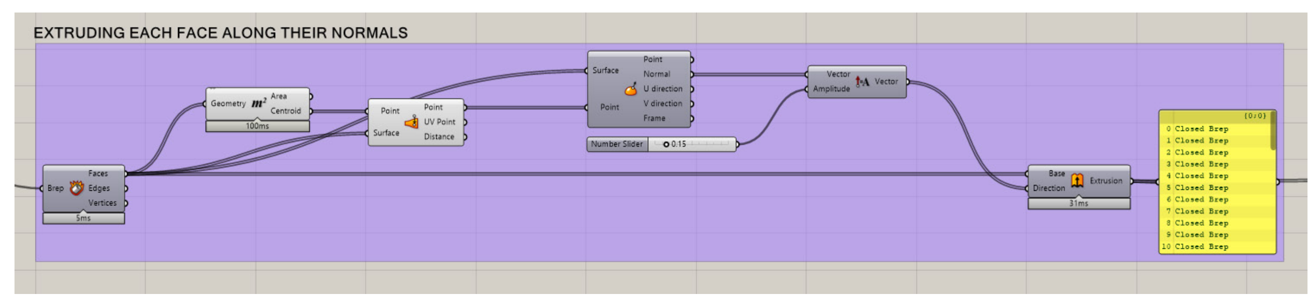
1. beginning polyline border



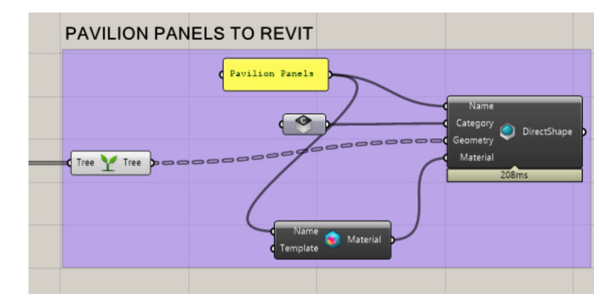
2. set curve



After modeling and sketching, I had a good idea of what moves to make and what components I need in grasshopper. My initial curve was derived from an underlay graphic from my studio work. This curve became the foundational piece of the pavilion that other components would build from. From there, I created a Brep that I could then extract edges from to make the structural pipes. Extracting the naked edges and interior edges separately allowed me to manipulate the structure of them independently from each other. This allowed me to give the structure for the naked edges more robust, while the structure for the interior edges were smaller. Structural connections between the pipes were needed, as the ends of the pipes were colliding, thus creating an issue for possible fabrication. To solve this, I implemented spherical joints at the vertecies of the structure, eliminating the collisions between pipes. After the structure was set, I moved on to giving the panels of the pavilion thickness. However, this proved to be tricky, as I had to establish the "normals" for each panel in order to extrude each panel in their respective perpendicular direction. Once the panels were given thickness, I used Rhino.Inside to send the panels into Revit as specigic components native to Revit.

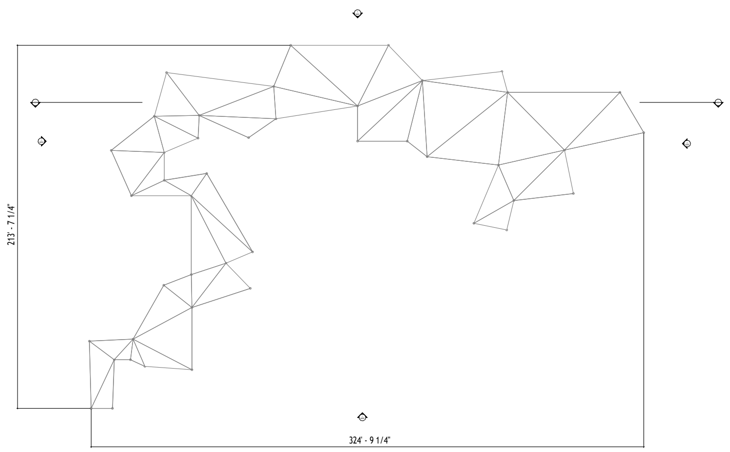


6. panel forming and extruding

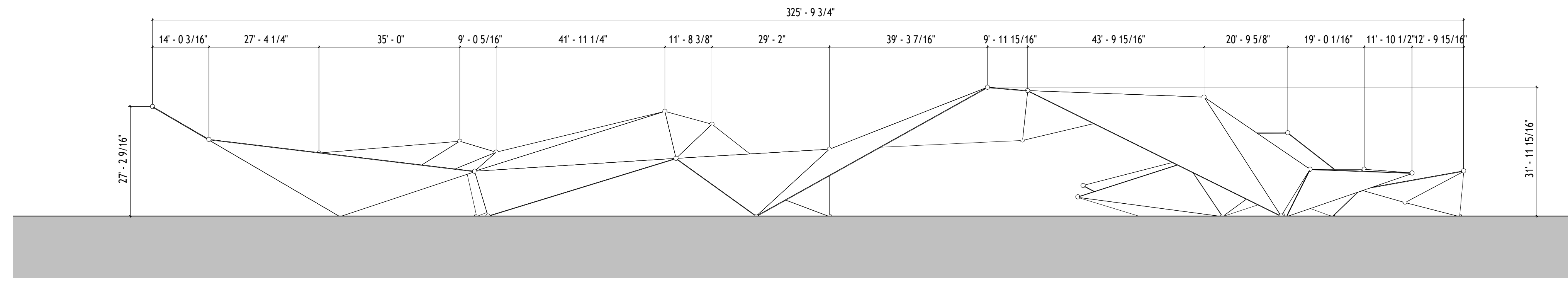


7. rhino inside

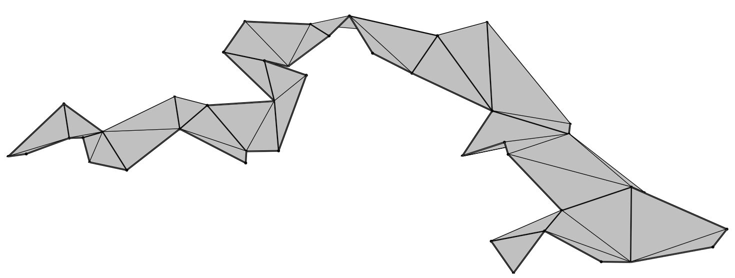
rhino + grasshopper



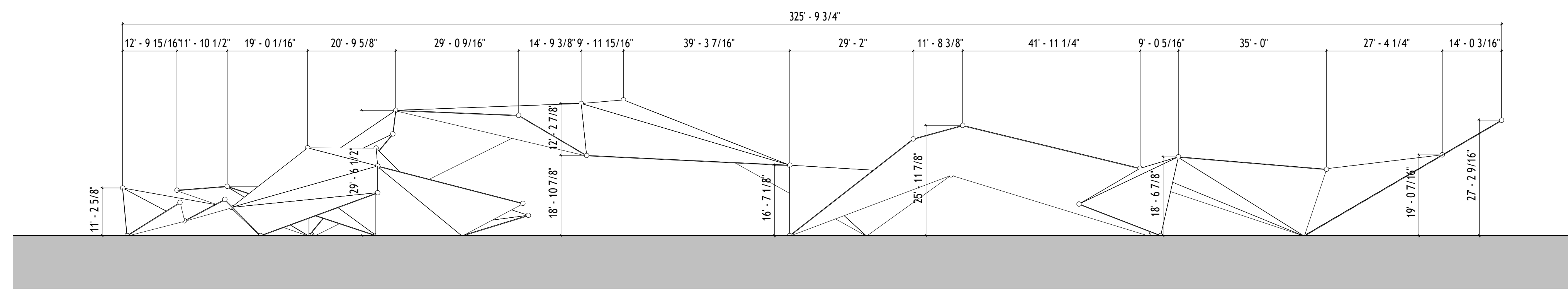
roof plan



north elevation



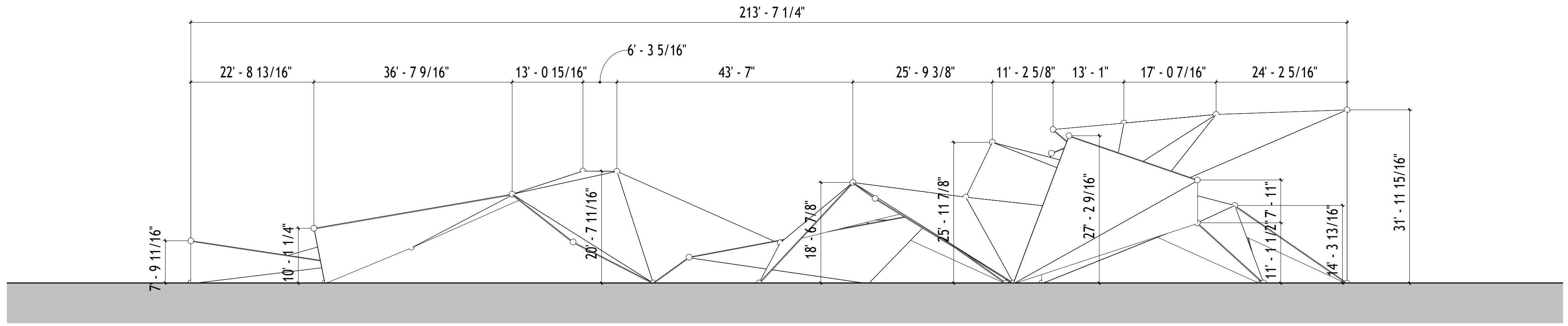
3D aerial view



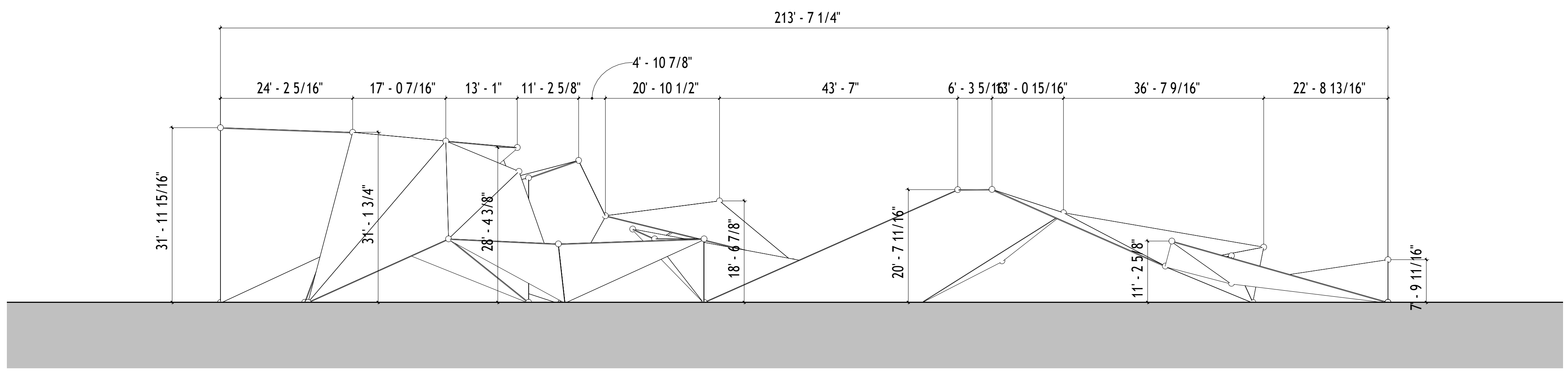
south elevation

In the Revit documentation stage, I realized this pavilion would be difficult to dimension properly. Grasshopper is a great tool for computational design, fabrication, and making certain processes efficient, but if not done correctly, it can make documentation difficult. While this was not the goal of this assignment, it is a good thing to learn and consider when designing, expecially if one gets into complex geometries and fabrication. Considering the complexity of the pavilion I designed, Rhino and Grasshopper were extremely effective and the perfect tools to use. This entire process made me understand how to effeiciently and effectively get complex geometries into Revit for documentation and material application, and then into Twinmotion for future rendering and development. This entire process is extremely useful for design, and can expedite the process even more when working ina professional setting.

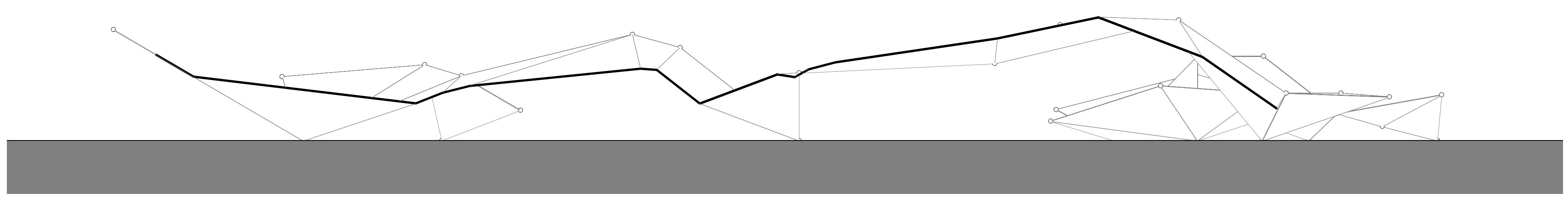
documentation



east elevation

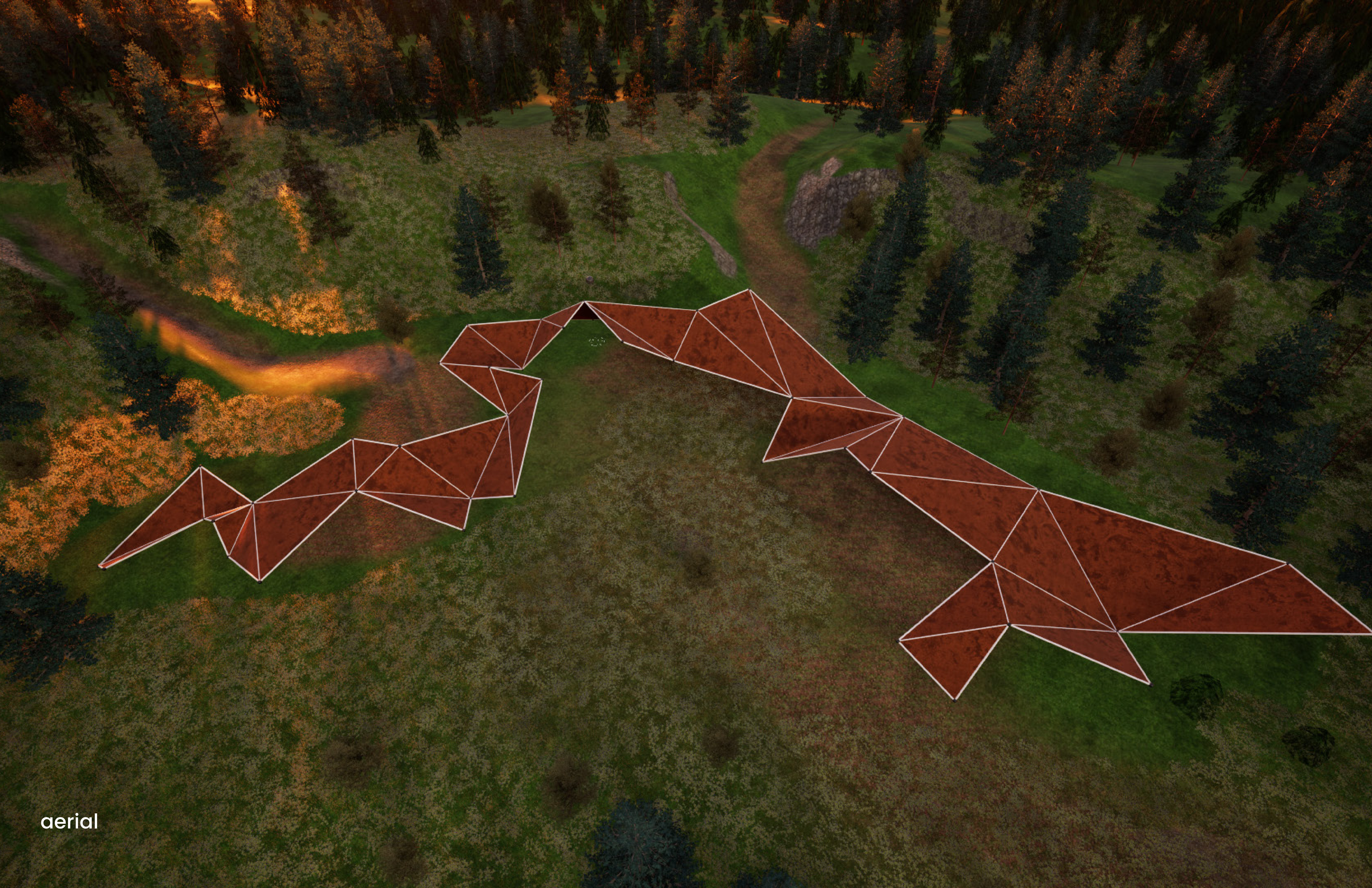


west elevation



section 1

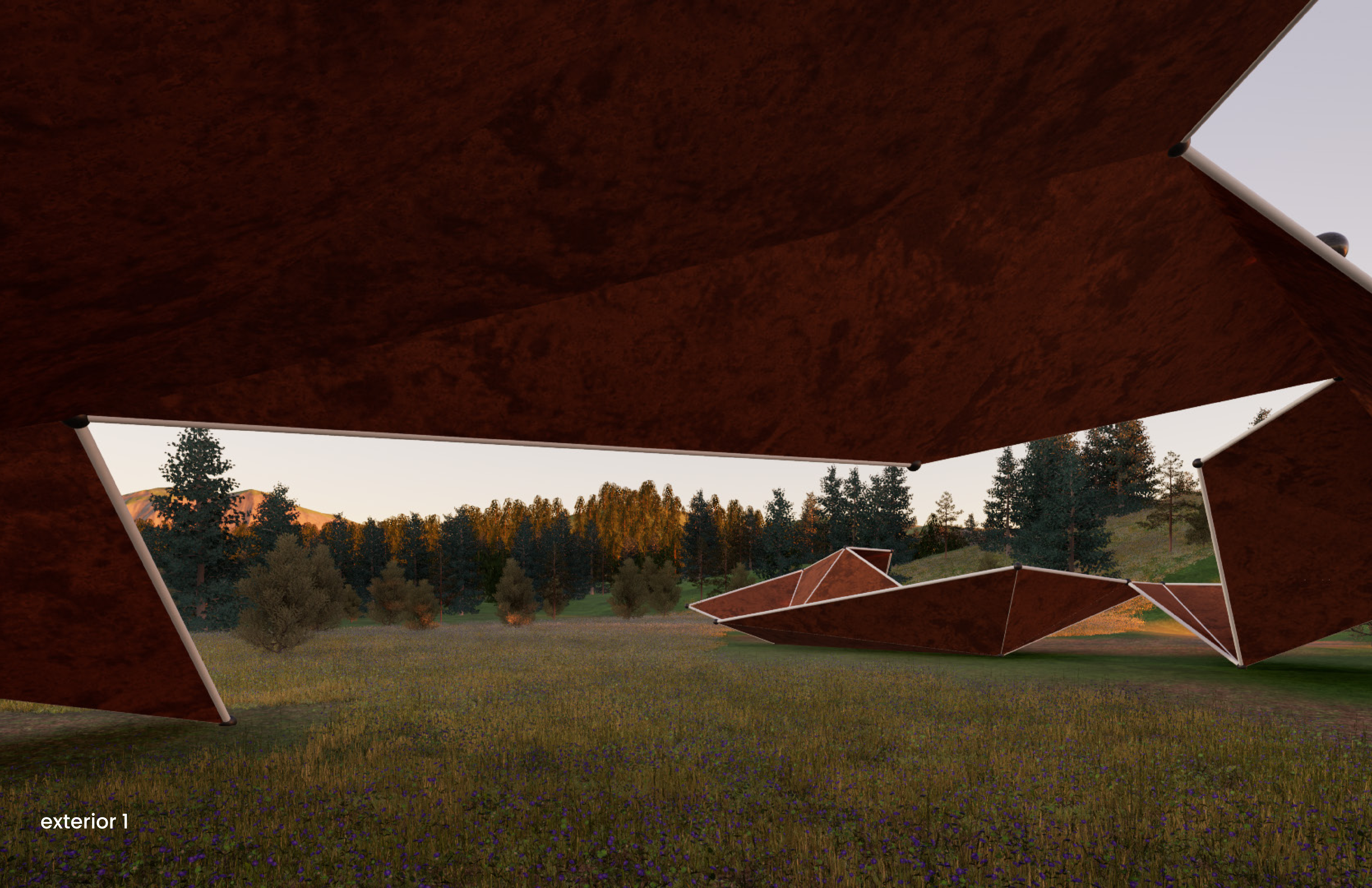
documentation



aerial

exterior 1


exterior 2



exterior 1